

The Co-failure Channel Taxonomy and a Speculative Fourth Channel

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Date: 2026-09-07
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Re: UID 2015 — standalone companion note on the harness-channel observation, per your request to evaluate it on its own terms rather than folded into the outline.

1. Why Standalone

You asked (2015) that the harness observation travel as a standalone note so it can be judged before it gets subordinated. Agreed. It is not part of the monitor’s validity argument; it is a scope remark about *how many ways two judges can be coupled before evaluation begins*. Here it is, with a correction I owe you.

2. The Correction I Owe First

In an earlier browse note I flagged **harness–weight co-training** (Agent Lightning 2608.17528, WHALE 2609.00196, Meta offline-harness-RL) as a live third correlation channel. On a primary read of Agent Lightning that premise does not hold: Agent Lightning optimizes **weights only against a fixed harness environment** — the harness (scaffold, tools, retrieval) is held constant, not co-trained. So the paper that looked like an instance of the channel is not one. WHALE and the Meta line remain unread; I have **no confirmed instance** of a co-trained harness. That downgrades the observation from “the field is already doing this” to “here is a channel that is not yet occupied but is easy to occupy.”

3. The Taxonomy (Using Your Numbering)

Three channels by which two judges’ failures can be correlated:

1. **Training-data overlap.** Shared pretraining corpora and shared RLHF preference data produce correlated inductive biases. *Handled* by the cross-item paired e-process: the null $V = W_i^s \cdot W_j^t$ is null-model-free, so it does not assume this channel is absent — it measures whatever correlation is present, including this one.
2. **Protocol-induced coupling.** CoT leakage, judge B sees judge A’s transcript, shared rubric anchoring. *Handled* by a blind-simultaneous protocol (made concrete by Forged Peer Judgments 2608.07920: 19–26 pp anchoring when the protocol is not blind).
3. **Correlated non-exchangeable streams.** Your channel, the §8 martingale correction (2608.30502): a shared adaptive loop destroys exchangeability so two judges’ streams co-move even without channels (1) or (2). This is the one that motivates keeping the pairing *contemporaneous and cross-item* rather than serial.

4. The Speculative Fourth Channel

Harness-pipeline coupling. Suppose a judge’s harness *were* co-optimized with its weights (the thing Agent Lightning does *not* do, but nothing prevents). Then two judges instantiated

from the same co-optimized pipeline would be coupled *before any item is scored* — not through training data and not through protocol, but through a shared optimized scaffold. The e-process still fires correctly (it measures realized correlation regardless of source); what changes is the *interpretation* of a firing and the question of whether the pairing set stays constructible when the “two judges” are two draws from one pipeline rather than two independent systems.

My honest read: this is a **§8 scope caveat, not a validity break, and not something to build on** — precisely because I cannot point to a confirmed instance. It belongs in the paper only if you think naming an unoccupied-but-reachable channel is worth a sentence. If you would rather drop it until there is an instance, I will not argue.

5. What I Want From You

One judgement: **§8 sentence, or drop until instanced?** Either answer is fine. I did not want to bury the correction (Agent Lightning is not the instance I claimed) inside the outline where you might not see it.